

These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**.
The purpose of this material is to make the topic clear and useful for students of Class 9
Computer Science.

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Algorithm Design Activities

Design and Evaluation Techniques

Design and evaluation techniques are used to create algorithms and check how well they work. They help us make step-by-step solutions to problems and then test them to see if they are fast, correct, and efficient.

We can check the efficiency of algorithms by two methods:

1. Time Complexity (how fast the algorithm runs)
2. Space Complexity (how much memory it uses)

Time Complexity

Time complexity tells us how much time an algorithm takes to solve a problem.

- If an algorithm takes less time, it is better.
- We usually measure time based on input size (like number of items).

If an algorithm becomes slow when input increases, it is not efficient.

Space Complexity

Space complexity tells us how much memory (space) an algorithm uses to solve a problem.

- If an algorithm uses less memory, it is better.
- We usually measure space based on input size and extra storage needed.

If an algorithm uses too much memory when input increases, it is not efficient.

Dry Run

A dry run means manually going through an algorithm step by step to see how it works, without using any tool or digital device.

Dry Run of a Flowchart

A dry run of a flowchart means we manually follow each step of the flowchart using sample input to see what output it gives.

Steps in Flowchart:

- Start
- Input Name
- Display Name
- Stop

Dry Run Example

Let’s assume input is: **Name = "Ali"**

Step	Flowchart Action	Value
1	Start	-
2	Input name	Name = Ali
3	Display name	Ali
4	Stop	-

Dry Run of a Pseudocode

A dry run of pseudocode means we manually follow each step of the pseudocode using sample input to see what output it gives.

Steps in Pseudocode:

START
INPUT Name
DISPLAY Name
STOP

Dry Run Example

Let’s assume input is: **Name = "Ali"**

Step	Pseudocode Action	Value
1	START	-
2	INPUT Name	Name = Ali
3	DISPLAY Name	Ali
4	STOP	-

Simulation

Simulation is the use of computer programs to create a model of a real-world process or system. This helps us understand how things work by testing different ideas or algorithms without needing to try them in real life.

It is also called computer simulation. It is also used to train people using artificial models of real-world situations.

Examples of Simulators:

- Flight simulator used to train pilots in a virtual airplane environment
- Driving simulator used to practice driving without real vehicles
- Weather simulator used to predict and study weather conditions
- Medical simulator used to train doctors for surgeries

Why Use Simulation?

1. Testing Algorithms:

We use simulation to test how an algorithm works with different data.

Example: We can test a sorting algorithm with small and large lists of numbers to see how fast it works.

2. Exploring Scenarios:

Simulation helps us try different situations without doing them in real life.

Example: We can test how plants grow with different amounts of water and sunlight.

Benefits of Simulation

- **Cost-Effective:**

Simulation is cheaper and faster than real experiments.

Example: Testing a new machine design on a computer is cheaper than building it in real life.

- **Safe:**

We can test dangerous situations without any risk.

Example: A fire simulation can help train people without starting a real fire.

- **Repeatable:**

We can run the same simulation many times with different settings.

Example: We can test weather conditions multiple times by changing temperature and rainfall.

Examples of Simulation

1. Weather Forecasting:

Meteorologists use simulation to predict the weather. They enter data like temperature, humidity, and wind speed into a computer model. The simulation shows how the weather may change in the next few days.

2. Traffic Flow:

City planners use simulation to study traffic. They test how cars move when they change roads or traffic lights. This helps them reduce traffic jams and improve road design.

LARP (Logic of Algorithms for Resolution of Problems)

LARP (Logic of Algorithms for Resolution of Problems) is a learning tool used to understand how algorithms work by running them step by step and seeing the results.

Simple Idea:

- It is a tool for learning algorithms
- It helps you test and run algorithms
- It shows step-by-step working and output

Why is LARP Important?

LARP helps you:

- Understand how algorithms work. For example, it can show an algorithm that checks whether a person should pay tax based on their annual salary. If the salary is above a certain limit, tax is applied; otherwise, no tax is charged.
- See how different inputs change the output.
- Practice writing and improving your own algorithms.

Error Identification and Debugging

When we write algorithms or create flowcharts in LARP, we sometimes make mistakes called errors or bugs. These mistakes can stop our algorithms from working correctly.

OR Error identification means finding mistakes (errors) in a flowchart, algorithm, or pseudocode.

Debugging is the process used to find and fix the errors so that the algorithm, flowchart or pseudocode work properly.

Types of Errors

There are three main types of errors you might encounter:

- **Syntax Errors:**

These happen when something is written incorrectly in an algorithm or flowchart. For example, missing a step or using the wrong symbol.

- **Runtime Errors:**

These happen while the algorithm is running. For example, trying to divide a number by zero.

- **Logical Errors:**

A logical error occurs due to wrong or poor logic used by the programmer.

For example, using the wrong condition in a decision step. So, the program runs but gives the wrong result.

Debugging Techniques

Debugging is the process of finding and fixing errors in an algorithm or flowchart.

Here are some common debugging techniques:

- **Trace the Steps:**

Go through each step of your algorithm or flowchart to find where it goes wrong.

- **Use Comments:**

Write notes or comments in your algorithm to explain what each part does. This helps in finding mistakes.

- **Check Conditions:**

Make sure all conditions in decision steps are correct.

- **Simplify the Problem:**

Break the algorithm into small parts and test each part separately.

Common Error Messages in LARP

Here are some common error messages you might see in LARP and what they mean:

- **Missing Step:**

You probably forgot to include an important step in your algorithm.

- **Undefined Variable:**

You are using a variable that has not been defined yet.

- **Invalid Operation:**

You are trying to perform an operation that is not allowed, such as dividing by zero.